

Left-handed spinor field: $\psi_a(x)$ $(\frac{1}{2}, 0)$ (Weyl)

$$\boxed{U(\Lambda)^{-1} \psi_a(x) U(\Lambda) = \rho_a^b(\Lambda) \psi_b(\Lambda^{-1}x)}$$

$$\rho_a^b(\Lambda) \rho_b^c(\Lambda') = \rho_a^c(\Lambda \Lambda')$$

$$\Lambda_{\mu}^{\nu} = \delta_{\mu}^{\nu} + \delta \omega_{\mu}^{\nu}$$

$$\rho_a^b(\Lambda) = \rho_a^b(\delta + \delta \omega) = \delta_a^b + \frac{i}{2} \delta \omega_{\mu\nu} (S_L^{\mu\nu})_a^b$$

where

$$(S_L^{\mu\nu})_a^b = - (S_L^{\nu\mu})_a^b$$

$$[S_L^{\mu\nu}, S_L^{\rho\sigma}] = i (g^{\mu\rho} S_L^{\nu\sigma} - (g^{\mu\sigma} S_L^{\nu\rho}) - (g^{\nu\rho} S_L^{\mu\sigma} - (g^{\nu\sigma} S_L^{\mu\rho}))$$

$$\bullet U(1 + \delta \omega) = \mathbb{I} + \frac{i}{2} \delta \omega_{\mu\nu} M^{\mu\nu}$$

$$\boxed{[\psi_a(x), M^{\mu\nu}] = \mathcal{L}^{\mu\nu} \psi_a(x) + (S_L^{\mu\nu})_a^b \psi_b}$$

$$\mathcal{L}^{\mu\nu} = \frac{i}{2} (x^{\mu} \partial^{\nu} - x^{\nu} \partial^{\mu})$$

• First term is also present for $n=0$, so we focus on the second term by evaluating at the origin $x^{\mu}=0$

$$M_{ij} = \epsilon_{ijk} J_k$$

$$[\psi_a(0), M^{ij}] = (S_L^{ij})_a^b \psi_b(0)$$

$$\epsilon^{ijk} [\psi_a(0), J_k] = (S_L^{ij})_a^b \psi_b(0)$$

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$\boxed{(S_L^{ij})_a^b = \frac{1}{2} \epsilon^{ijk} \sigma_k^b_a}$$

• Once we have J_i we can recover K_k (Boosts)

$$K_k = M^{k0} \text{ from}$$

$$J_k = M_k + M_k^{\dagger}, \quad K_k = i(M_k - M_k^{\dagger})$$

$$[\psi_a(0), J_k] = \frac{1}{2} \sigma_k \psi_a(0)$$

$$[\psi_a(0), M_k] + [\psi_a(0), M_k^{\dagger}] = \frac{1}{2} \sigma_k \psi_a(0)$$

as for $(\frac{1}{2}, 0)$ we find

$$J_k = (N_k + N_k^\dagger) |(\frac{1}{2}, 0)\rangle = (n + n') |(\frac{1}{2}, 0)\rangle = \frac{1}{2}$$

$$\text{so } N_k^\dagger |(\frac{1}{2}, 0)\rangle = 0$$

$$\text{and so } K_k = iN_k = iJ_k = M^{40}$$

$$(S^{k0})_\alpha^\beta = \frac{1}{2} i(\sigma^k)_\alpha^\beta$$

Consider the Herm. conjugate of $\psi_\alpha(x)$ which swaps the reps $\rightarrow \overline{SU(2)}$

$$[\psi_\alpha(x)]^\dagger = \psi_{\dot{\alpha}}^\dagger(x) \quad (\text{Right handed spinor})$$

$$U(\Lambda)^{-1} \psi_{\dot{\alpha}}^\dagger(x) U(\Lambda) = R_{\dot{\alpha}}^{\dot{\beta}}(\Lambda) \psi_{\dot{\beta}}^\dagger(\Lambda^{-1}x)$$

$$[\psi_{\dot{\alpha}}^\dagger(0), M^{\mu\nu}] = (S_R^{\mu\nu})_{\dot{\alpha}}^{\dot{\beta}} \psi_{\dot{\beta}}^\dagger(0)$$

(We have again ignored the $\mathcal{L}^{\mu\nu}$ contribution)

Taking the Herm. conj. of this eq.

$$[M^{\mu\nu}, \psi_\alpha(0)] = [(S_L^{\mu\nu})_\alpha^\beta]^\dagger \psi_\beta(0)$$

and so

$$[(S_R^{\mu\nu})_{\dot{\alpha}}^{\dot{\beta}}] = -[(S_L^{\mu\nu})_\alpha^\beta]^\dagger$$

* Let us look at Lorentz transformations of higher order spinors: Take $C_{\alpha\beta}(x)$ which carries two $(\frac{1}{2}, 0)$ indices

Lorentz:

$$U(\Lambda)^{-1} C_{\alpha\beta}(x) U(\Lambda) = \rho_\alpha^\gamma(\Lambda) \rho_\beta^\delta(\Lambda) C_{\gamma\delta}(\Lambda^{-1}x)$$

Recall that $\alpha, \beta \in \{1, 2\}$

Q: Can we find the irreducible irreps for $C_{\alpha\beta}$ under Lorentz?

Two spin $(\frac{1}{2})$ particles can have spin sums of

$$|S| = \underbrace{+1}_{\text{Vector}}, \underbrace{0}_{\text{Scalar}}$$

$$\psi_\alpha \chi_B \quad \sum_{\alpha_1 \dots \alpha_p} = \sum_{\alpha_1} \sum_{\alpha_2} \dots \sum_{\alpha_p} \quad \epsilon_{\beta_1 \dots \beta_n} = -p! \delta_{\beta_1 \dots \beta_n}^{\alpha_1 \dots \alpha_n}$$

$$\begin{aligned} \epsilon^{\alpha\beta} \epsilon_{\delta\gamma} &= -2! \delta_{\delta\gamma}^{\alpha\beta} \\ &= -2 \delta_{\delta}^{\alpha} \delta_{\gamma}^{\beta} = (-\delta_{\delta}^{\alpha} \delta_{\gamma}^{\beta} + \delta_{\delta}^{\beta} \delta_{\gamma}^{\alpha}) \\ \epsilon^{\alpha\beta} \epsilon_{\delta\gamma} &= \delta_{\delta}^{\beta} \delta_{\gamma}^{\alpha} - \delta_{\delta}^{\alpha} \delta_{\gamma}^{\beta} \end{aligned}$$

$$\text{Let } B = \delta$$

$$\begin{aligned} \epsilon^{\alpha\beta} \epsilon_{\beta\gamma} &= 2 \delta_{\gamma}^{\alpha} - \delta_{\beta}^{\alpha} \delta_{\gamma}^{\beta} = 2 \delta_{\gamma}^{\alpha} - \delta_{\gamma}^{\alpha} \\ &= \delta_{\gamma}^{\alpha} \end{aligned}$$

as one would find in the text

$$\underline{2} \text{ of } SU(2)$$

$$\underline{2} \otimes \underline{2} = \underline{1}_A \oplus \underline{3}_S$$

$$(\frac{1}{2}, 0) \oplus (\frac{1}{2}, 0) = (0, 0)_A \oplus (1, 0)$$

$$\epsilon_{\alpha\beta} \rightarrow \rho_{\alpha}^{\gamma} \epsilon_{\gamma\delta} \rho_{\beta}^{\delta}$$

$$\det(A) \epsilon_{\alpha\beta} = \epsilon_{\gamma\delta} A_{\alpha}^{\gamma} A_{\beta}^{\delta}$$

Thus

$$\epsilon_{\alpha\beta} \mapsto \rho_{\alpha}^{\gamma} \epsilon_{\gamma\delta} \rho_{\beta}^{\delta} = \epsilon_{\alpha\beta} \det(\rho)$$

$$\text{Since } \rho: SU(2) \Rightarrow \det(\rho) = 1$$

$$\epsilon_{\alpha\beta} \mapsto \epsilon_{\alpha\beta}$$

i.e. The Lorentz group acts trivially on $\epsilon_{\alpha\beta}$ and is the "SU(2) metric"

$$\rho_{\alpha}^{\gamma}(\Lambda) \epsilon_{\gamma\delta} \rho_{\beta}^{\delta}(\Lambda) = \epsilon_{\alpha\beta}$$

$$\text{just as with } g_{\mu\nu} \Lambda_{\mu}^{\alpha} \Lambda_{\nu}^{\beta} = g_{\alpha\beta}$$

$$\epsilon_{\alpha\beta} \text{ " " } (0, 0) = \underline{1} \text{ of } SU(2)$$

$$\square = \psi_{\alpha}$$

$$\square \otimes \square = \square + \square$$

$$\underline{2} \otimes \underline{2} = \underline{1} + \underline{3}$$

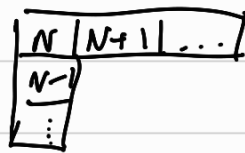
$$\square = \psi_{\alpha}$$

$$\psi_{\alpha} \psi_{\beta} \dots \psi_{\gamma} = \psi^*$$

$$\left\{ \begin{array}{c} \square \\ \square \\ \square \end{array} \right\} N-1$$

Complex (adj. rep ψ_a =

dim of rep:



hook

hook:

(# of Boxes to the right) $\oplus$ (# of boxes below)

ex: hook $\left(\begin{array}{|c|c|c|} \hline 6 & 4 & 2 \\ \hline 4 & 2 & \\ \hline 3 & 1 & \\ \hline 2 & & \\ \hline \end{array} \right) = 6 \cdot 4 \cdot 4 \cdot 2 \cdot 3 \cdot 1$ $\oplus$ Box

* Thus

$$\square \otimes \square = \square \oplus \square$$

$\underline{2} \quad \underline{2} \quad \quad \underline{3} \quad \quad \underline{1}$

$$\dim \square = \frac{2 \cdot 3}{2} = 3$$

$$\dim \square = 2/2 = 1$$

Thus, $\psi_a \chi_b = \epsilon_{ab} D(x) + \underbrace{G_{ab}(x)}_{\text{symm.}}$

* The same is true for $SU(2)$
 $(0, 1/2) \otimes (0, 1/2) = (0, 0) \oplus (0, 1)$
 which has the invariant "metric"
 $\epsilon_{\alpha\beta}$

Now look at

$$(1/2, 0) \otimes (0, 1/2)$$

$$\square \otimes \bar{\square} =$$

$$\psi_a \bar{\chi}_{\dot{a}} = \frac{1}{2} (\psi \sigma_{\mu} \bar{\chi}) \sigma^{\mu}_{\alpha\dot{\alpha}}$$

$$\Rightarrow V_{\mu} = \frac{1}{2} (\psi \sigma_{\mu} \bar{\chi})$$

$$A_{\alpha\dot{\alpha}}(x)$$

$$\left(SL(2, \mathbb{C}) \right) \cong SU(2) \times \bar{SU}(2)$$

Thus: $V_{\alpha i}(x) = V_m \sigma^m_{\alpha i}$

$\sigma^m_{\alpha i}$: Also an invariant symbol

$$\sigma^m_{\alpha i} = (1, \vec{\sigma})$$

* Important Note:

We can deduce an invariant object (0,0) or 1 from a given tensoring from (0,0) appearing $\swarrow g_{\mu\nu}$

ex:

$$\left[\begin{array}{ccc} (1/2, 1/2) & \otimes & (1/2, 1/2) \\ V^{\mu} & \otimes & V^{\mu} \end{array} \right] = (0,0) \oplus (0,1) \oplus (1,0) \oplus (1,1)$$

Recall: $B^{\mu\nu}(x) = A^{\mu\nu} + \underbrace{S^{\mu\nu}}_{\text{Symm/traceless}} + \frac{1}{4} g^{\mu\nu} M^{\alpha\lambda} B_{\alpha\lambda}$

$T(x) \equiv M^{\alpha\lambda} B_{\alpha\lambda}(x)$ is (0,0)

$S^{\mu\nu} \equiv B^{(\mu\nu)} - \frac{1}{4} g^{\mu\nu} g^{\alpha\lambda} B_{\alpha\lambda}$

(1,1) is $S^{\mu\nu}$

$S^{\mu\nu}$: $n \times n$:

$$\left[\frac{n^2 - n}{2} - 1 + n \right]$$

For $n=4$, $5+4=9$

$S^{\mu\nu} = 9$

$T(x) = 1$

Note: The dim of the rep is equal to $(2n+1)(2n'+1)$

From Sred. book we simply multiply the two numbers in (A,B)

$A^{\mu\nu} = B^{(\mu\nu)}$ must break down into

$$(1,0) \oplus (0,1)$$

• Recall,

$$\psi_\alpha \otimes \chi_\beta = \frac{1}{2} \epsilon_{\alpha\beta} (\psi\chi) + \frac{1}{2} (\sigma^{\mu\nu} \epsilon^T)_{\alpha\beta} (\psi \sigma_{\mu\nu} \chi)$$

$(1,0)$: three-vector / space of self dual tensors of rank two

• Contains $(\alpha\beta)$

Can use the generators $S_L^{\mu\nu}$ and $S_R^{\mu\nu}$

$$\text{Since } \text{Tr}(\sigma) = 0$$

$$(S_L^{\mu\nu})_\alpha{}^\alpha = 0$$

The $SL(2, \mathbb{C})$ trace is w.r.t. ϵ

$$\epsilon^{ab} (S_L^{\mu\nu})_{ab} = 0$$

Thus, $(S_L^{\mu\nu})_{\alpha\beta}$ must be symmetric in $\alpha\beta$

$$(S_L^{10})_\alpha{}^\beta = i (S_L^{23})_\alpha{}^\beta$$

$$(S_L^{\mu\nu})_\alpha{}^\beta = -\frac{i}{2} \epsilon^{\mu\nu\rho\sigma} (S_{L\rho\sigma})_\alpha{}^\beta \quad (1,0)$$

(Self-dual)

$$S^{\mu\nu} = * S^{\mu\nu}$$

$$G^{\mu\nu}(x) = (S_L^{\mu\nu})^{\alpha\beta} G_{\alpha\beta}(x)$$

$(0,1)$

$$G^{\dagger\mu\nu}(x) = -(S_L^{\mu\nu})^{\dot{\alpha}\dot{\beta}} G_{\dot{\alpha}\dot{\beta}}^\dagger(x)$$

(anti-self dual)

$$G^{\mu\nu}(x) = -\frac{i}{2} \epsilon^{\mu\nu\rho\sigma} G_{\rho\sigma}(x) \quad (1,0)$$

$$G^{\dagger\mu\nu}(x) = i/2 \epsilon^{\mu\nu\rho\sigma} G_{\rho\sigma}^\dagger(x) \quad (0,1)$$

$$G^{\mu\nu}(x) = \frac{1}{2} (A^{\mu\nu}(x) + * A^{\mu\nu}(x)) \quad (1,0)$$

$$G^{\dagger\mu\nu}(x) = \frac{1}{2} (A^{\mu\nu}(x) - * A^{\mu\nu}(x)) \quad (0,1)$$

